CHAPTER 1

# Partial Derivatives and Gradients

This chapter will introduce you to partial derivatives and gradients, equipping you with the tools to study functions of multiple variables. We will explore how these concepts provide valuable insights into optimization, vector calculus, and various fields of science and engineering.

Partial derivatives come into play when dealing with functions that depend on multiple variables. Unlike ordinary derivatives that consider changes along a single variable, partial derivatives focus on how a function changes concerning each individual variable while holding the others constant. In essence, partial derivatives measure the rate of change of a function with respect to one variable, while keeping the other variables fixed.

The notation for a partial derivative of a function $f(x, y, \ldots)$ with respect to a specific variable, say $x$, is denoted as $\frac{\partial f}{\partial x}$. Similarly, $\frac{\partial f}{\partial y}$ represents the partial derivative with respect to $y$, and so on. It is essential to remember that when taking partial derivatives, we treat the other variables as constants during the differentiation process.

The gradient is a vector that combines the partial derivatives of a function. It provides a concise representation of the direction and magnitude of the steepest ascent or descent of the function. The gradient vector points in the direction of the greatest rate of increase of the function. By understanding the gradient, we gain insights into optimizing functions and finding critical points where the function reaches maximum or minimum values.

Throughout this chapter, we will explore the following key topics related to partial derivatives and gradients:

- Calculating partial derivatives: We will delve into the techniques and rules for computing partial derivatives of various functions, including polynomials, exponential functions, and trigonometric functions. We will also explore higher-order partial derivatives and mixed partial derivatives.
- Interpreting partial derivatives: Understanding the geometric and physical interpretations of partial derivatives is essential. We will discuss the notion of tangent planes, directional derivatives, and the relationship between partial derivatives and local linearity.
- Gradient vectors and their properties: We will introduce this concept, including it connection to the direction of steepest ascent, its relationship with partial derivatives,

and how it relates to level curves and level surfaces.

- Applications of partial derivatives and gradients: We will explore various applications of these concepts, including optimization problems, constrained optimization, tangent planes, linear approximations, and their relevance in fields like physics, economics, and engineering.

By grasping the concepts of partial derivatives and gradients, you will unlock a powerful mathematical framework for analyzing and optimizing functions of multiple variables. These tools will equip you to tackle advanced calculus problems and gain deeper insights into the behavior of functions in diverse fields.

## 1.1 Calculating Partial Derivatives

For a function of two variables, $f(x, y)$, we can take the derivative with respect to $x$ or with respect to $y$. These are called the *partial derivatives* of $f$. Formally, the partial derivatives are defined as:

**Limit Definition of Partial Derivatives**

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Let's consider a polynomial function of two variables: $f(x, y) = 3x^2 + y^3 + 4xy$. We will use the limit definition to find the partial derivative with respect to $x$, then compare this to what we already know about derivatives of single-variable functions. Recall that if we can describe a function as a sum of two other functions, the derivative of the original function is the same as the sum of the derivatives of the other functions. That is,

$$\text{if } f(x) = g(x) + h(x)$$

$$\text{then } f'(x) = g'(x) + h'(x)$$

Let's then define $r(x, y) = 3x^2$, $s(x, y) = y^3$, and $t(x, y) = 4xy$. And so $f(x, y) = r(x, y) + s(x, y) + t(x, y)$, which means $f_x(x, y) = r_x(x, y) + s_x(x, y) + t_x(x, y)$. Then,

$$\begin{aligned} f_x(x, y) &= \lim_{h \to 0} \frac{r(x + h, y) - r(x, y)}{h} + \lim_{h \to 0} \frac{s(x + h, y) - s(x, y)}{h} + \lim_{h \to 0} \frac{t(x + h, y) - t(x, y)}{h} \\ &= \lim_{h \to 0} \frac{3(x + h)^2 - 3x^2}{h} + \lim_{h \to 0} \frac{y^3 - y^3}{h} + \lim_{h \to 0} \frac{4(x + h)y - 4xy}{h} \\ &= \lim_{h \to 0} \frac{3x^2 + 6xh + h^2 - 3x^2}{h} + 0 + \lim_{h \to 0} \frac{4xy + 4hy - 4xy}{h} \end{aligned}$$

Notice that $s_x(x, y) = 0$. This term only had $y$, and its derivative with respect to $x$ is zero. Continuing,

$$\begin{aligned} f_x(x, y) &= \lim_{h \to 0} \frac{6xh + h^2}{h} + \lim_{h \to 0} \frac{4hy}{h} = \lim_{h \to 0} 6x + h + \lim_{h \to 0} 4y \\ &= 6x + 4y \end{aligned}$$

As you can see, $r_x(x, y) = 6x$ and $t_x(x, y) = 4y$. Recall the polynomial rule for single derivatives. The derivative of $3x^2$ is $6x$, which is also what we see with the partial derivative in this case. What about the other term, $4xy$? Well, we know the derivative of $bx$, where $b$ is a constant, is $b$. The partial derivative of $4xy$ with respect to $x$ being $4y$ suggests the rule for determining partial derivatives:

**Rule for Finding Partial Derivatives of $f(x, y)$**

1. To find the partial derivative with respect to $x$, $f_x$, treat $y$ as a constant and differentiate with respect to $x$.
2. To find the partial derivative with respect to $y$, $f_y$, treat $x$ as a constant and differentiate with respect to $y$.

Let's check this by predicting $f_y$, then using the limit definition to confirm our prediction. Applying the polynomial rule, we predict that $f_y$ is:

$$f_y(x, y) = 3y^2 + 4x$$

Which we found by treating $x$ as a constant and taking the derivative of each term with respect to $y$. Let's see if we get the same result using the limit definition of the derivative with respect to $y$:

$$\begin{aligned} f_y(x, y) &= \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h} \\ &= \lim_{h \to 0} \frac{\left[3x^2 + (y + h)^3 + 4x(y + h)\right] - \left[3x^2 + y^3 + 4xy\right]}{h} \\ &= \lim_{h \to 0} \frac{3x^2 + y^3 + 3y^2h + 3yh^2 + h^3 + 4xy + 4xh - 3x^2 - y^3 - 4xy}{h} \\ &= \lim_{h \to 0} \frac{3y^2h + 3yh^2 + h^3 + 4xh}{h} = \lim_{h \to 0} 3y^2 + 3yh + h^2 + 4x = 3y^2 + 4x \end{aligned}$$

Which is our expected result. In summary, you find the partial derivative with respect to a particular variable by treating all the other variables as constants and differentiating with respect to the particular variable, applying the rules of differentiation you've already learned.

### 1.1.1 Partial Derivative Notation

There are many ways to denote a partial derivative. We've already seen one way, $f_x$ and $f_y$. Another common notation uses a lowercase Greek letter delta, and a further uses capital D. They are shown below:

**Partial Derivative Notations**

$$f_x(x,y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,y) = D_x f$$

$$f_y(x,y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,y) = D_y f$$

## *Exercise 1* First Partial Derivatives

Find $f_x$ and $f_y$ for the following functions.

*Working Space*

1. $f(x,y) = 3x^4 + 4x^2y^3$
2. $f(x,y) = xe^{-y}$
3. $f(x,y) = \sqrt{3x + 4y^2}$
4. $f(x,y) = \sin x^2 y$
5. $f(x,y) = \ln(x^y)$

*Answer on Page 35*

### 1.1.2 Partial Derivatives of Functions of More than Two Variables

The above method of determining partial derivatives applies to functions with three, four, or any number of variables.

**Example**: Find all the first derivatives of the function $f(x, y, z) = y \cos\left(x^2 + 3z\right)$.

**Solution**:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[y \cos\left(x^2 + 3z\right)\right] = -y \sin\left(x^2 + 3z\right)\left(\frac{\partial}{\partial x}\left(x^2 + 3z\right)\right)$$

$$\frac{\partial f}{\partial x} = -2xy \sin\left(x^2 + 3z\right)$$

And

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[y \cos\left(x^2 + 3z\right)\right]$$

$$\frac{\partial f}{\partial y} = \cos\left(x^2 + 3z\right)$$

And

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}\left[y \cos\left(x^2 + 3z\right)\right] = -y \sin\left(x^2 + 3z\right)\left(\frac{\partial}{\partial z}\left(x^2 + 3z\right)\right)$$

$$\frac{\partial f}{\partial z} = -3y \sin\left(x^2 + 3z\right)$$

## *Exercise 2* Partial Derivatives with 3 or More Variables

Find all first partial derivatives of the following functions.

*Working Space*

1. $f = \sin\left(x^2 - y^2\right)\cos\left(\sqrt{z}\right)$
2. $q = \sqrt[3]{t^3 + u^3 \sin(5v)}$
3. $w = x^z y^x$

*Answer on Page 35*

### 1.1.3 Higher Order Partial Derivatives

Just like with single-variable equations, we can take the partial derivative more than once. There are also several notations for second partial derivatives.

**Second Partial Derivative Notation**

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

Notice that for $\left(\partial^2 f/\partial y \partial x\right)$, we first take the derivative with respect to $x$, then with respect to $y$.

**Example**: Find all the second order partial derivatives of $f(x, y) = 2x^2 - x^3y^2 + y^3$.

**Solution**: We begin by finding $f_x$ and $f_y$:

$$f_x(x, y) = 4x - 3x^2y^2$$

$$f_y(x, y) = -2x^3y + 3y^2$$

We then take another partial derivative to find all the second order partial derivatives:

$$f_{xx}(x, y) = \frac{\partial}{\partial x} f_x(x, y) = \frac{\partial}{\partial x}\left(4x - 3x^2y^2\right) = 4 - 6xy^2$$

$$f_{xy}(x, y) = \frac{\partial}{\partial y} f_x(x, y) = \frac{\partial}{\partial y}\left(4x - 3x^2y^2\right) = -6x^2y$$

$$f_{yx}(x, y) = \frac{\partial}{\partial x} f_y(x, y) = \frac{\partial}{\partial x}\left(-2x^3y + 3y^2\right) = -6x^2y$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y} f_y(x, y) = \frac{\partial}{\partial y}\left(-2x^3y + 3y^2\right) = -2x^3 + 6y$$

What do you notice about $f_{xy}$ and $f_{yx}$? They are the same! This is not a coincidence of the particular function used in the example. For most functions, $f_{xy} = f_{yx}$, as stated by Clairaut's theorem.

**Clairaut's Theorem**

If $f$ is defined on a disk $D$ and $f_{xy}$ and $f_{yx}$ are both continuous on $D$, then $f_{xy} = f_{yx}$ on $D$.

This is also true for third, fourth, and higher-order derivatives.

## *Exercise 3* Clairaut's Theorem

Show that Clairaut's theorem holds for the following functions (show that $f_{xy} = f_{yx}$).

1. $f(x, y) = e^{2xy} \sin x$
2. $f(x, y) = \frac{x^2}{x+y}$
3. $f(x, y) = \ln(2x + 3y)$

*Working Space*

*Answer on Page 37*

## *Exercise 4* Second Order Partial Derivatives

Find all second order partial derivatives of the function.

*Working Space*

1. $f(x, y) = x^5y^2 - 3x^3y^2$
2. $v = \sin\left(p^3 + q^2\right)$
3. $T = e^{-3r}\cos\theta^2$

*Answer on Page 37*

### 1.1.4 The Chain Rule

For single-variable functions, where $y = f(x)$ and $x = g(t)$, we have seen that:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Which is the Chain Rule for single-variable functions. For multi-variable functions, there are several versions of the Chain Rule, depending on how the variables and functions are defined. First, we consider the case where $z = f(x, y)$ and $x = g(t)$ and $y = h(t)$ (i.e. $f$ is a multi-variable function of $x$ and $y$, while $x$ and $y$ are single-variable functions of $t$).

This means that $z$ is an indirect function of $t$:

$$z = f(x, y) = f\left(g(t), h(t)\right)$$

Then the derivative of $z$ with respect to $t$ is given by:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

**Example**: If $z = xy^2 + 3x^4y$, where $x = 2\sin(t)$ and $y = \cos(3t)$, find $dz/dt$ when $t = \pi/2$.

**Solution**: First, we apply the Chain Rule to $z$:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial x}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}\left[xy^2 + 3x^4y\right] \cdot \frac{d}{dt}\left[2\sin(t)\right] + \frac{\partial}{\partial y}\left[xy^2 + 3x^4y\right] \cdot \frac{d}{dt}\left[\cos(3t)\right]$$

$$= \left(y^2 + 12x^3y\right) \cdot \left(2\cos(t)\right) + \left(2xy + 3x^4\right) \cdot \left(-3\sin(3t)\right)$$

When $t = \pi/2$, $\cos(t) = 0$, $\sin(3t) = -1$, $x = 2$, and $y = 0$. Substituting:

$$\frac{dz}{dt} = (0 + 0) \cdot (0) + \left(0 + 3(2)^4\right) \cdot (-3 \cdot -1)$$

$$= 3(2)^4 \cdot 3 = 144$$

Another case is where $x$ and $y$ are also multi-variable functions. Consider $z = f(x, y)$, $x = g(s, t)$, and $y = h(s, t)$. This means $z$ is an indirect function of $s$ and $t$:

$$z = f(x, y) = f\left(g(s, t), h(s, t)\right)$$

In this case, there are two partial derivatives of $z$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

**Example**: Find $\partial z/\partial s$ and $\partial z/\partial t$ if $z = e^{2x}\cos y$, $x = s^2t$, and $y = st^2$.

**Solution**: First, let's find $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial s}\left[s^2 t\right] + \frac{\partial}{\partial y}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial s}\left[st^2\right]$$
$$= \left(2e^{2x}\cos y\right) \cdot (2st) + \left(-e^{2x}\sin y\right) \cdot \left(t^2\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = 4ste^{2s^2 t}\cos\left(st^2\right) - t^2 e^{2s^2 t}\sin\left(st^2\right)$$

And finding $\partial z/\partial t$:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$
$$= \frac{\partial}{\partial x}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial t}\left[s^2 t\right] + \frac{\partial}{\partial y}\left[e^{2x}\cos y\right] \cdot \frac{\partial}{\partial t}\left[st^2\right]$$
$$= \left(2e^{2x}\cos y\right) \cdot \left(s^2\right) + \left(-e^{2x}\sin y\right) \cdot (2st)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = 2s^2 e^{2s^2 t}\cos\left(st^2\right) - 2kte^{2s^2 t}\sin\left(st^2\right)$$

## *Exercise 5* The Chain Rule for Multivariable Functions

Find $dz/dt$ or $\partial z/\partial s$ and $\partial z/\partial t$.

*Working Space*

1. $z = \sin x \cos y$, $x = 3\sqrt{t}$, $y = 2/t$
2. $z = \sqrt{1 + xy}$, $x = \tan t$, $y = \arctan t$
3. $z = \arctan\left(x^2 + y^2\right)$, $x = t \ln s$, $y = se^t$
4. $z = \sqrt{x}e^{xy}$, $x = 1 + st$, $y = s^2 - t^2$

*Answer on Page 38*

## 1.2 Interpreting Partial Derivatives

What is the meaning of a partial derivative? Recall that $z = f(x, y)$ plots a surface, $S$. Consider the function $z = \cos y - x^2$, shown in figure 1.1.

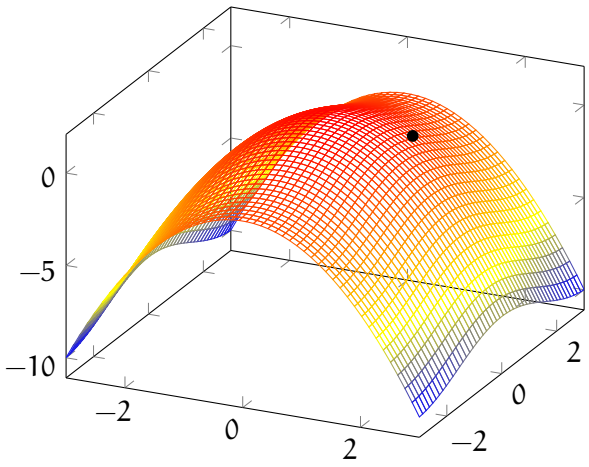



Figure 1.1: The surface $z = \cos y - x^2$

We can see that $f(1, \pi/3) = -1/2$; therefore, the point $(1, \pi/3, -1/2)$ lies on the surface $z = \cos y - x^2$ (the black dot shown in figure **??**). If we fix $y$ such that $y = \pi/3$, we are looking at the intersection between the surface and the plane $y = \pi/3$ (see figure **??**).

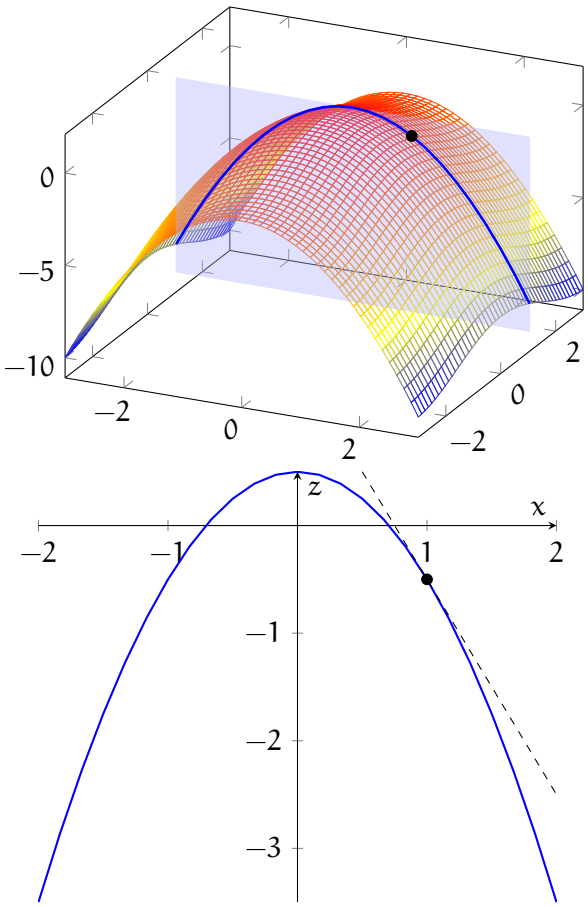



Figure 1.2: The intersection between the surface $z = \cos y - x^2$ and $y = \pi/3$ is the parabola $z(x) = 1/2 - x^2$

We can describe this intersection as $g(x) = f(x, \pi/3)$, so the slope of a tangent line to this intersection is given by $g'(x) = f_x(x, \pi/3)$. This means, geometrically, $f_x(1, \pi/3)$ is the slope of the line that lies tangent to $z = f(x, y)$ at the point $(1, \pi/3, -1/2)$ and in the plane $y = \pi/3$ (see figure 1.2). Alternatively, you could think of $f_x$ as the slope of the tangent line to the surface that is parallel to the x-axis.

Similarly, we can fix $x = 1$ and look at the intersection between the surface $z = \cos y - x^2$ and the plane $x = 1$ (see figure 1.3. Just like before, we can describe this intersection as $h(y) = f(1, y)$, which means the slope of a line tangent to the intersection is given by $h'(y) = f_y(1, y)$. Therefore, as with $f_x$, $f_y(a, b)$ gives the slope of a line tangent to the point $(a, b, f(a, b))$ and parallel to the y-axis.

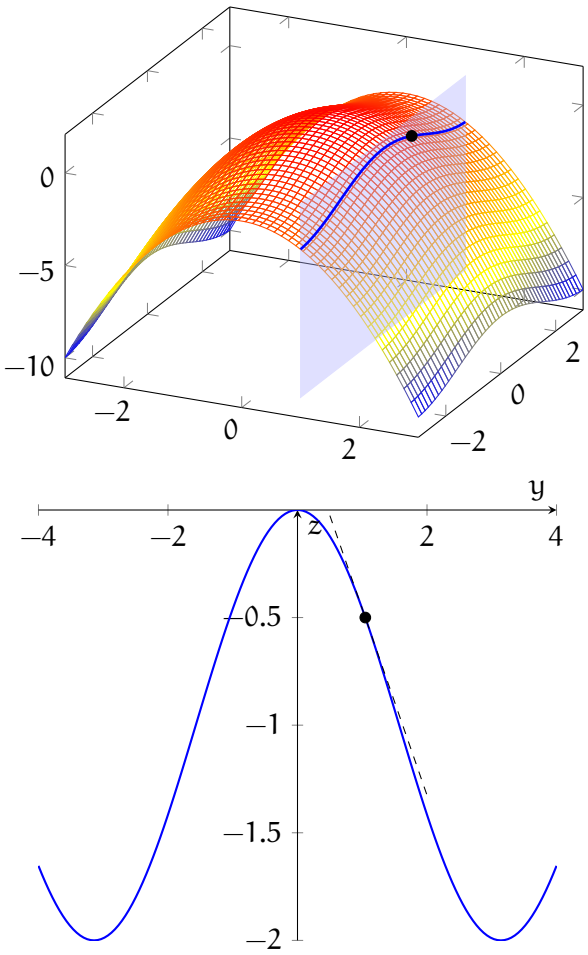



Figure 1.3: The intersection between the surface $z = \cos y - x^2$ and $x = 1$ is the trigonometric function $z = \cos y - 1$

**Example**: The density of bacterial growth at a point $(x, y)$ on a flat agar plate is given by $D = 45/\left(2 + x^2 + y^2\right)$. Find the rate of change of bacterial density at the point $(1, 3)$ (a) in the x-direction and (b) in the y-direction. Interpret the meaning of your results.

**Solution**: The rate of change of a two-variable function in the x-direction is given by the partial derivative with respect to x:

$$D_x = \frac{\partial}{\partial x}\frac{45}{2 + x^2 + y^2} = \frac{-45\left(\partial/\partial x\right)\left(2 + x^2 + y^2\right)}{\left(2 + x^2 + y^2\right)^2}$$

$$= \frac{-90x}{\left(2 + x^2 + y^2\right)^2}$$

The rate of change in the x-direction at $(x, y) = (1, 3)$ is given by:

$$D_x(1,3) = \frac{-90(1)}{(2+1^2+3^2)^2} = \frac{-90}{(12)^2} = \frac{-90}{144} = -\frac{5}{8}$$

This means that at $(1, 3)$, the density of bacteria is decreasing as you move away $x = 0$ along the line $y = 3$.

Similarly, the rate of change in the y-direction is given by the partial derivative with respect to $y$:

$$D_y = \frac{\partial}{\partial y}\frac{45}{2+x^2+y^2} = \frac{-45\,(\partial/\partial y)\left(2+x^2+y^2\right)}{(2+x^2+y^2)^2}$$
$$= \frac{-90y}{(2+x^2+y^2)^2}$$

The rate of change in the y-direction at $(x, y) = (1, 3)$ is given by:

$$D_y(1,3) = \frac{-90(3)}{(2+1^2+3^2)^2} = \frac{-270}{144} = -\frac{15}{8}$$

This means that at $(1, 3)$ the density of bacteria is decreasing faster along the y-direction than along the x-direction.

### *Exercise 6* Using partial derivatives to find tangent lines

Find equations for tangent lines to the surface at the given $xy$-coordinate. In which direction is the function changing the fastest?

1. $z = x^2 e^{y/x}$, $(1, -1)$
2. $z = \cos x + y \sin y$, $(\pi, \pi/2)$
3. $z = x^2 y - 3xy^2$, $(3, 2)$

*Working Space*

*Answer on Page 40*

## 1.3 Gradient Vectors

The gradient vector is used to find the direction of the maximum rate of change of a surface (for example, the steepest part of a mountain). In order to understand the gradient, we must first discuss directional derivatives. Recall that the partial derivatives, $f_x$ and $f_y$, can be used to define a plane tangent to the surface $z = f(x, y)$ (see figure 1.4). Directional derivatives allow us to find the slope of the tangent plane in directions other than the $x$- and $y$-directions.

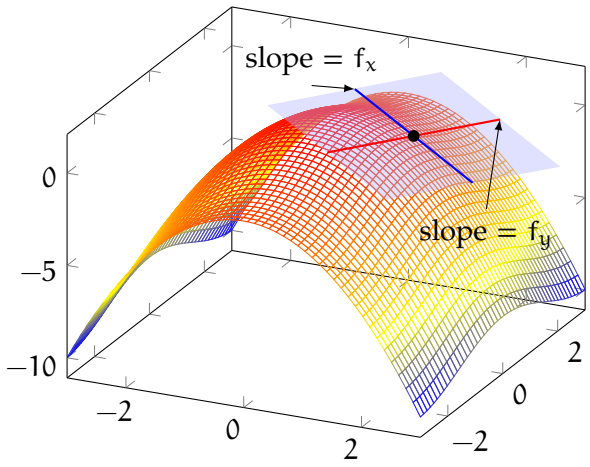



Figure 1.4: The directional derivatives, $f_x$ and $f_y$ define a tangent plane

### 1.3.1 Directional Derivatives

The contour map in figure 1.5 shows the elevation, $f(x, y)$ for a mountain. You already know that you can use the partial derivatives, $f_x$ and $f_y$ to find the rate of change in elevation going east-west or north-south. But what about other directions? Suppose the hiking path you're on goes north-east. How can you predict the steepness (i.e. the rate of elevation change) along this path? The directional derivative allows us to find the rate of change in any direction.

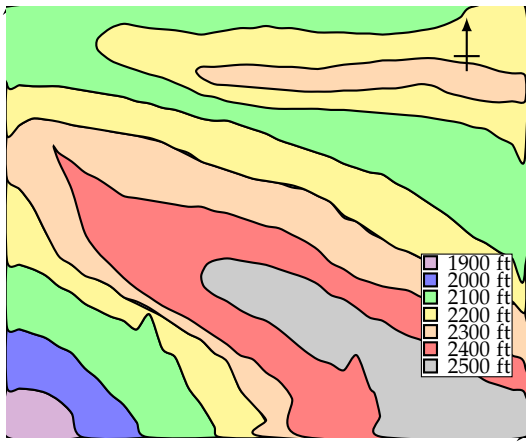



Figure 1.5: The contour plot shows the elevation of a mountain. $f_x$ gives the slope going east, while $f_y$ gives the slope going north

At some point, $(x_o, y_o)$, the partial derivatives $f_x(x_o, y_o)$ and $f_y(x_o, y_o)$ give the rate of change of elevation in the east-west and north-south directions, respectively (see figure 1.6).

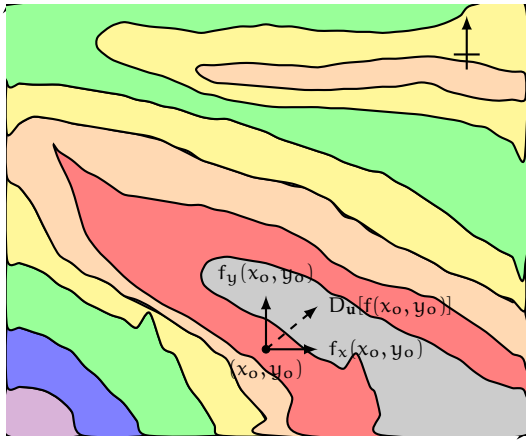



Figure 1.6: If $\mathbf{u}$ points north-east, then the directional derivative of $f(x, y)$ at $(x_o, y_o)$, $D_{\mathbf{u}}[f(x_o, y_o)]$, tells the rate of change going north-east

To find the rate of change at $(x_o, y_o)$, in the direction of some arbitrary unit vector, $\mathbf{u} = [a, b] = a\mathbf{i} + b\mathbf{j}$, we first note that the point $Q = (x_o, y_o, z_o)$, where $z_o = f(x_o, y_o)$, lies on the surface defined by $z = f(x, y)$. There is a vertical plane, P, that passes through Q and points in the direction of $\mathbf{u}$. This intersection defines curve C, which lies on the surface, and the slope of this curve at $Q = (x_o, y_o, z_o)$ is the directional derivative of H in the direction of $\mathbf{u}$ (see figure 1.7).

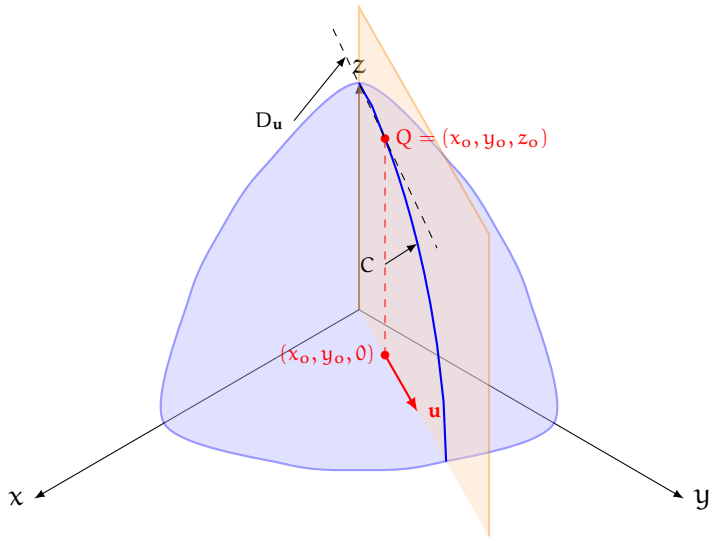



Figure 1.7: The slope of the curve formed between the plane parallel to $\mathbf{u}$ and the surface $z = f(x, y)$ is the directional derivative, $D_{\mathbf{u}}$

We can choose another point, $R = (x, y, z)$, the is $h$ units away from Q along $\mathbf{u}$ (see 1.8). Then the change in $x$ is $x - x_o = ha$ and the change is $y$ is $y - y_o = hb$. And the slope

from Q to R is given by:

$$\frac{\delta z}{h} = \frac{f(x,y) - f(x_o,y_o)}{h}$$

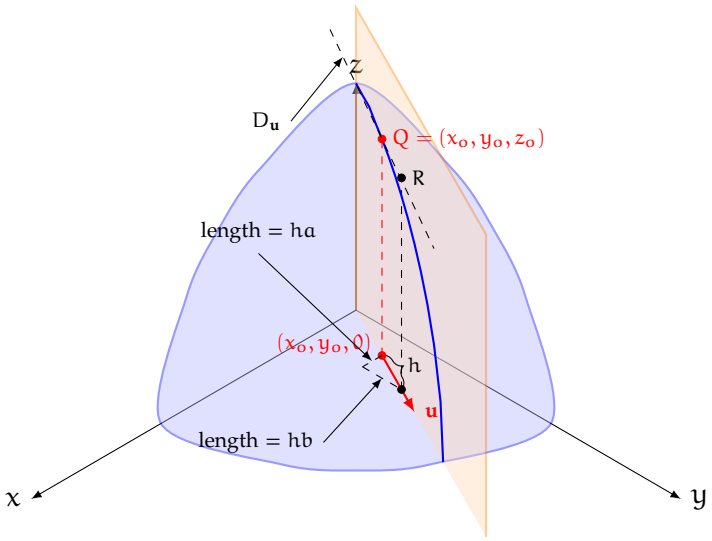



Figure 1.8: A second point, R, along **u** is h units away along **u**

We find the directional derivative by substituting for x and y and taking the limit as h goes to zero:

$$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_o + ha, y_o + hb) - f(x_o, y_o)}{h}$$

How is this related to $f_x$ and $f_y$? Let's define $g(h)$ such that $g(h) = f(x_o + ha, y_o + hb)$. Then

$$g'(0) = \lim_{h \to 0} \frac{g(h) - g(0)}{h}$$

$$= \lim_{h \to 0} = \frac{f(x_o + ha, y_o + hb) - f(x_o, y_o)}{h} = D_{\mathbf{u}} f(x_o, y_o)$$

We can also apply the Chain Rule to $g(h)$:

$$g'(h) = \frac{\partial f}{\partial x}\frac{dx}{dh} + \frac{\partial f}{\partial y}\frac{dy}{dh} = f_x(x,y)a + f_y(x,y)b$$

Substituting $h = 0$, $x = x_o$, and $y = y_o$, we see that:

$$g'(0) = f_x(x_o, y_o)a + f_y(x_o, y_o)b$$

Which means that:

$$D_{\mathbf{u}} f(x_o, y_o) = f_x(x_o, y_o)a + f_y(x_o, y_o)b$$

So a directional derivative is:

**The Directional Derivative**

Let $f$ be a differentiable function and $\mathbf{u}$ be a unit vector, $\mathbf{u} = [a, b]$. Then the directional derivative in the direction of $\mathbf{u}$ is:

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)a + f_y(x, y)b = \mathbf{u}_x \left[\frac{\partial}{\partial x} f(x, y)\right] + \mathbf{u}_y \left[\frac{\partial}{\partial y} f(x, y)\right] \tag{1.1}$$

Where $\mathbf{u}_x$ and $\mathbf{u}_y$ are the $x$- and $y$-components of $\mathbf{u}$, respectively.

**Example**: Find the directional derivative $D_{\mathbf{u}} f(x, y)$ if $f(x, y) = y^3 - 3xy + 4x^2$ and $\mathbf{u}$ is the unit vector given by the angle $\theta = \pi/3$. What is the rate of change in the direction of $\mathbf{u}$ at $(1, 2)$?

**Solution**: We can describe $\mathbf{u}$ thusly:

$$\mathbf{u} = \left[\cos\frac{\pi}{3}, \sin\frac{\pi}{3}\right] = \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]$$

And therefore:

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)\left(\frac{1}{2}\right) + f_y(x, y)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\partial}{\partial x}\left(y^3 - 3xy + 4x^2\right)\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left(y^3 - 3xy + 4x^2\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{2}(-3y + 8x) + \frac{\sqrt{3}}{2}\left(3y^2 - 3x\right)$$

$$= \frac{-3}{2}y + 4x + \frac{3\sqrt{3}}{2}y^2 - \frac{3\sqrt{3}}{2}x = \frac{3\sqrt{3}}{2}y^2 + \frac{8 - 3\sqrt{3}}{2}x - \frac{3}{2}y$$

And therefore $D_{\mathbf{u}} f(1, 2)$ is:

$$= \frac{3\sqrt{3}}{2}(2)^2 + \frac{8 - 3\sqrt{3}}{2}(1) - \frac{3}{2}(2) = 6\sqrt{3} + 4 - \frac{3\sqrt{3}}{2} - 3$$

$$= 1 + \frac{9\sqrt{3}}{2}$$

### 1.3.2 Unit Vectors in Two Dimensions

What if the given vector is not a unit vector? We can scale the given vector to find a unit vector in the same direction:

**Example**: Find the directional derivative of $f(x,y) = 3x\sqrt{y}$ at $(1,4)$ in the direction of $\mathbf{v} = [2,1]$.

**Solution**: First, we need to find a unit vector in the same direction as $\mathbf{v}$. There are several ways to do this. In two dimensions, a unit vector in the same direction as $\mathbf{v}$ can be found using trigonometry (see figure 1.9 for an illustration).

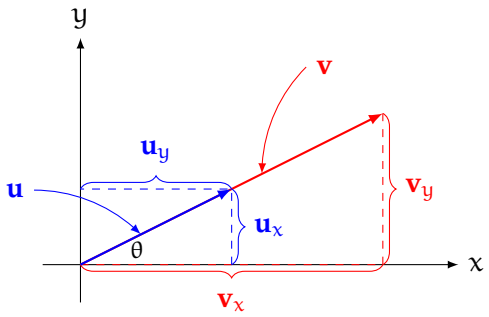



Figure 1.9: $\mathbf{u}$ is a unit vector in the same direction as $\mathbf{v}$

We know that $\theta = \arctan(\mathbf{v}_y/\mathbf{v}_x)$. Therefore, the x-component of the unit vector, $\mathbf{u}$, is given by:

$$\mathbf{u}_x = |\mathbf{u}|\cos\theta = \cos\left(\arctan\frac{\mathbf{v}_y}{\mathbf{v}_x}\right)$$

Similarly, we know that:

$$\mathbf{u}_y = |\mathbf{u}|\sin\theta = \sin\left(\arctan\frac{\mathbf{v}_y}{\mathbf{v}_x}\right)$$

(Recall that since $\mathbf{u}$ is a unit vector, $|\mathbf{u}| = 1$).

Let's use this method to find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{v} = [2,1]$:

$$\mathbf{u}_x = \cos\left(\arctan\frac{1}{2}\right) \approx \cos(0.464) = \frac{2}{\sqrt{5}}$$

$$\mathbf{u}_y = \sin\left(\arctan\frac{1}{2}\right) \approx \sin(0.464) = \frac{1}{\sqrt{5}}$$

Therefore, a unit vector in the same direction as $\mathbf{v}$ is $\mathbf{u} = \left[2/\sqrt{5}, 1/\sqrt{5}\right]$.

And we can find the directional derivative:

$$D_{\mathbf{u}}(x,y) = \mathbf{u}_x\left[\frac{\partial}{\partial x}f(x,y)\right] + \mathbf{u}_y\left[\frac{\partial}{\partial y}f(x,y)\right]$$

$$D_{\mathbf{u}}(x,y) = \left(\frac{2}{\sqrt{5}}\right)\left[\frac{\partial}{\partial x}\left(3x\sqrt{y}\right)\right] + \left(\frac{1}{\sqrt{5}}\right)\left[\frac{\partial}{\partial y}\left(3x\sqrt{y}\right)\right]$$
$$D_{\mathbf{u}}(x,y) = \left(\frac{2}{\sqrt{5}}\right)\left(3\sqrt{y}\right) + \left(\frac{1}{\sqrt{5}}\right)\left(\frac{3x}{2\sqrt{y}}\right)$$
$$D_{\mathbf{u}}(x,y) = \frac{12y + 3x}{2\sqrt{5y}}$$

To find the magnitude of the directional derivative at $(1,4)$, we substitute for $x$ and $y$:

$$D_{\mathbf{u}}(1,4) = \frac{12(4) + 3(1)}{2\sqrt{5(4)}} = \frac{51}{4\sqrt{5}} \approx 5.702$$

### 1.3.3 Unit Vectors in Higher Dimensions

The trigonometric explanation for finding unit vectors is more difficult to visualize in higher dimensions. However, there is another method that works well in 2, 3, and higher dimensions. Recall that the magnitude of a vector, $\mathbf{v} = [\mathbf{v}_x, \mathbf{v}_y]$ is given by $|\mathbf{v}| = \sqrt{(\mathbf{v}_x)^2 + (\mathbf{v}_y)^2}$. For a vector with $n$ dimensions, $\mathbf{v} = [\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n]$, the magnitude is given by $|\mathbf{v}| = \sqrt{(\mathbf{v}_1)^2 + (\mathbf{v}_2)^2 + \cdots + (\mathbf{v}_n)^2}$.

To find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{v}$, we can scale $\mathbf{v}$ up or down so that its magnitude is 1. We can do this by dividing by $\mathbf{v}$'s magnitude. Consider the two-dimensional vector used in the last example, $\mathbf{v} = [2,1]$. Its magnitude is:

$$|\mathbf{v}| = \sqrt{(2)^2 + (1)^2} = \sqrt{5}$$

Let's check if $\mathbf{v}/|\mathbf{v}|$ is a unit vector:

$$\frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{1}{\sqrt{5}}\right)[2,1] = \left[\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right]$$

And the magnitude of this scaled vector is:

$$\left|\frac{\mathbf{v}}{|\mathbf{v}|}\right| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{1} = 1$$

Notice our unit vector is the same as we found using the trigonometric method above.

Another way to think of the question is: what factor, $k$, can we multiply $\mathbf{v}$ by to yield a vector with a magnitude of 1? Let's see this method for the 3-dimensional vector $\mathbf{v} = [3,2,1]$. We are looking for a $k$ such that:

$$|k\mathbf{v}| = 1$$

$$|k\mathbf{v}| = |[3k, 2k, 1k]| = \sqrt{(3k)^2 + (2k)^2 + (1k)^2}$$
$$= \sqrt{9k^2 + 4k^2 + k^2} = k\sqrt{14} = 1$$

Which implies that $k = 1/\sqrt{14}$, which is $1/|\mathbf{v}|$. And therefore a unit vector in the same direction as $\mathbf{v} = [3, 2, 1]$ is:

$$\mathbf{u} = \frac{1}{\sqrt{14}}[3, 2, 1] = \left[\frac{3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}\right]$$

## *Exercise 7* Finding Directional Derivatives

Find the directional derivative of the function at the given point in the direction of the given vector.

1. $f(x, y) = e^{3x} \sin 2y$, $(0, \pi/6)$, $\mathbf{v} = [-3, 4]$
2. $f(x, y) = x^2y + xy^3$, $(2, 4)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$
3. $f(x, y, z) = \ln\left(x^2 + 3y - z\right)$, $(2, 2, 1)$, $\mathbf{v} = [1, 1, 1]$

*Working Space*

*Answer on Page 41*

### 1.3.4 Maximizing the Gradient

The directional derivative can be written as the dot product of two vectors:

$$D_u f(x,y) = a f_x(x,y) + b f_y(x,y) = [f_x(x,y), f_y(x,y)] \cdot \mathbf{u}$$

The first vector, $[f_x(x,y), f_y(x,y)]$, is called *the gradient of $f$*, and is noted as $\nabla f$.

**The Gradient**

For a two-variable function, $f(x,y)$, the gradient of $f$ is the vector:

$$\nabla f = [f_x(x,y), f_y(x,y)] = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$$

Where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the $x$- and $y$-directions, respectively.

Think back to the elevation example we opened the chapter with. What if we wanted to complete our ascent as quickly as possible? We would want to know the direction in which the elevation is changing the fastest. This occurs when the direction we are going is the same direction as the gradient vector, $\nabla f$.

Recall that the dot product is defined as:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta$$

Where $\theta$ is the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$. Applying this to the directional derivative, we see that:

$$D_u f = \nabla f \cdot \mathbf{u} = |\nabla f|\,|\mathbf{u}| \cos\theta = |\nabla f| \cos\theta$$

Which is at its maximum when $\nabla f$ and $\mathbf{u}$ point in the same direction (because $\cos(0) = 1$). Therefore, the gradient vector points in the direction of maximum change and the magnitude of that vector is the rate of maximum change.

**Example**: Find the maximum rate of change of $f(x,y) = 4y\sqrt{x}$ at $(4,1)$. In what direction does the maximum change occur?

**Solution**: We begin by finding $\nabla f$:

$$\nabla f = \left[\frac{\partial}{\partial x}\left(4y\sqrt{x}\right), \frac{\partial}{\partial y}\left(4y\sqrt{x}\right)\right]$$

$$\nabla f = \left[\frac{2y}{\sqrt{x}}, 4\sqrt{x}\right]$$

And thus,

$$\nabla f(4,1) = \left[\frac{2(1)}{\sqrt{4}}, 4\sqrt{4}\right] = [1,8]$$

Therefore, the maximum value of $\nabla f$ at $(4,1)$ is:

$$|\nabla f| = \sqrt{1^2 + 8^2} = \sqrt{65}$$

in the direction of the vector $[1,8]$.

## *Exercise 8* Using the Gradient to find Maximum Change

Suppose you are climbing a mountain whose elevation is described by $z = 3000 - 0.01x^2 - 0.02y^2$. Take the positive $x$-direction to be east and the positive $y$-direction to be north.

1. If you are at $(x, y) = (50, 50)$, what is your elevation?
2. If you walk south, will you ascend or descend?
3. If you walk northwest, will you ascend or descend? Will the rate of elevation change be greater or less than if you walked south?
4. In what direction should you walk for the steepest ascent? What will your ascension rate be?

*Working Space*

*Answer on Page 43*

### 1.3.5 Conclusion

in summary, we have learned that the gradient of some multivariable function is the point of steepest increase of the function, represented as:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rangle$$

The directional derivative is the gradient at some point $a$ dotted by the unit vector:

$$D_{\mathbf{u}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}$$

It is best to think of the gradient as an arrow pointing in the steepest uphill direction (vector), and the directional derivative as the slope of any specific uphill direction (scalar).

## 1.4 Applications of Partial Derivatives and Gradients

### 1.4.1 Laplace's Equation

A partial differential equation that has applications in fluid dynamics and electronics is Laplace's Equation. Solutions to Laplace's Equation are called *harmonic functions*.

**Laplace's Equation**

Consider a twice-differentiable function, $f$. In two dimensions, Laplace's Equation is given by:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

And in three dimensions,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Another way to represent Laplace's Equation is:

$$\delta f = \nabla^2 f = \nabla \cdot \nabla f = 0$$

Where $\nabla^2 = \delta$ is called the *Laplace operator*.

**Example**: Determine whether or not $f = x^2 + y^2$ is a solution to Laplace's Equation.

**Solution**: We are checking to see if $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$ for $f(x, y) = x^2 + y^2$. Finding $\partial^2 f / \partial x^2$:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^2 + y^2\right)\right]$$

$$= \frac{\partial}{\partial x}(2x) = 2$$

And finding $\partial^2 f/\partial y^2$:

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^2 + y^2\right)\right]$$
$$= \frac{\partial}{\partial y}(2x) = 2$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2 + 2 = 4 \neq 0$. Therefore, $f(x, y) = x^2 + y^2$ is not a solution to Laplace's Equation.

## *Exercise 9* Solutions to Laplace's Equation

Determine whether the function is a solution to Laplace's Equation.

*Working Space*

1. $f(x, y) = x^2 - y^2$
2. $f(x, y) = \sin x \cosh y + \cos x \sinh y$
3. $f(x, y) = e^{-x}\cos y - e^{-y}\cos x$

*Answer on Page 44*

### 1.4.2 The Wave Equation

Another useful equation with partial derivatives is the Wave Equation:

$$\frac{\partial^2 f}{\partial t^2} = a^2 \frac{\partial^2 f}{\partial x^2}$$

Where $f$ is a function of $x$ and $t$ and $a$ is a constant. This equation describes waves, such as a vibrating string, light waves, or sound waves.

**Example**: Show that $f(x,t) = \sin(x - at)$ satisfies the Wave Equation.

**Solution**: First, we find the second partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}(\sin(x - at))\right] = \frac{\partial}{\partial t}[-a\cos(x - at)] = -a^2 \sin(x - at)$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(\sin(x - at))\right] = \frac{\partial}{\partial x}[\cos(x - at)] = -\sin(x - at)$$

And we see that:

$$a^2 \frac{\partial^2 f}{\partial x^2} = -a^2 \sin(x - at) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, this function satisfies the Wave Equation.

## *Exercise 10* The Wave Equation

Show that the following functions satisfy the Wave Equation:

*Working Space*

1. $f(x,t) = \cos(kx)\cos(\alpha kt)$
2. $f(x,t) = \sin(x-\alpha t) + \ln(x+\alpha t)$
3. $f(x,t) = \frac{t}{\alpha^2 t^2 - x^2}$

*Answer on Page 45*

### 1.4.3 Cobb-Douglas Production Function

The Cobb-Douglas function describes the marginal utility of capital and labor as theorized by the economists Charles Cobb and Paul Douglas. Capital investments are things like new machinery, expanded factories, or raw materials. Labor investments involve hiring more workers or improving working conditions to improve work rates. We can describe total production, P, as a function of labor, L, and capital, K. Cobb and Douglas posit three conditions:

1. Without either labor or capital, production will cease.
2. The marginal utility of labor is proportional to the amount of production per unit of labor.
3. The marginal utility of capital is proportional to the amount of production per unit of capital.

The marginal utility of labor is given by the partial derivative, $\partial P/\partial L$ and the production per unit of labor is given by $P/L$. Therefore, statement 2 says that:

$$\frac{\partial P}{\partial L} = \alpha \frac{P}{L}$$

where $\alpha$ is some constant. Keeping K constant at $K = K_0$, we have the differential equation:

$$\frac{dP}{dL} = \alpha \frac{P}{L}$$

Solving, we find that:

$$P(L, K_0) = C_1(K_0) L^{\alpha}$$

We make $C_1$ a function of $K_0$ because it could depend on $K_0$. In a similar manner to above, we can write statement 3 as a mathematical statement:

$$\frac{\partial P}{\partial K} = \beta \frac{P}{K}$$

where $\beta$ is also a constant. Keeping $L = L_0$ and solving, we see that:

$$P(L_0, K) = C_2(L_0) K^{\beta}$$

again, we assume $C_2$ is a function of the fixed labor, $L_0$. Combining these equations, we get:

$$P(L, K) = bL^{\alpha}K^{\beta}$$

where $b$ is a constant independent of capital and labor. Additionally, from statement 1, we know that $\alpha > 0$ and $\beta > 0$. What happens if both labor and capital are increased by a factor of $n$? Let's examine the effect on P:

$$P(nL, nK) = b(nL)^{\alpha}(nK)^{\beta}$$

$$P(nL, nK) = n^{\alpha+\beta} bL^{\alpha} K^{beta} = n^{\alpha+\beta} P(L, K)$$

Cobb and Douglas noted that if $\alpha + \beta = 1$, then $P(nL, nK) = nP(L, K)$, and therefore increasing labor and capital by a factor of $n$ increases production by a factor of $n$ as well. Therefore, the Cobb-Douglas equation assumes $\alpha + \beta = 1$ and can be written as:

$$P(L, K) = bL^{\alpha}K^{1-\alpha}$$

## *Exercise 11* Cobb-Douglas Production Model

Cobb and Douglas modeled production in the US from 1900 to 1922 with the equation $P(L, K) = 1.01L^{0.75}K^{0.75}$.

1. Express the marginal utility of labor as a function of L and K.
2. Express the marginal utility of capital as a function of L and K.
3. In 1916, $L = 382$ and $K = 126$ (compared to initial values of 100 in 1900). What is the marginal utility of labor in 1916? Of capital?
4. Based on your answer to the previous question, would you invest in capital or labor if you owned a factory in 1916? Why?

*Working Space*

*Answer on Page 47*

Appendix A

# Answers to Exercises

## Answer to Exercise 1 (on page 4)

1. $f_x(x,y) = \frac{\partial}{\partial x}\left[3x^4 + 4x^2y^3\right] = 12x^3 + 8y^3$ and $f_y(x,y) = \frac{\partial}{\partial y}\left[3x^4 + 4x^2y^3\right] = 12x^2y^2$
2. $f_x(x,y) = \frac{\partial}{\partial x}\left(xe^{-y}\right) = e^{-y}$ and $f_y(x,y) = \frac{\partial}{\partial y}\left(xe^{-y}\right) = -xe^{-y}$
3. $f_x(x,y) = \frac{\partial}{\partial x}\sqrt{3x+4y^2} = \left(\frac{1}{2\sqrt{3x+4y^2}}\right)\left(\frac{\partial}{\partial x}\left(3x+4y^2\right)\right) = \frac{3}{2\sqrt{3x+4y^2}}$ and $f)y(x,y) = \frac{\partial}{\partial y}\sqrt{3x+4y^2} = \frac{1}{2\sqrt{3x+4y^2}}\left(\frac{\partial}{\partial y}\left(3x+4y^2\right)\right) = \frac{8y}{2\sqrt{3x+4y^2}} = \frac{4y}{\sqrt{3x+4y^2}}$
4. $f_x(x,y) = \frac{\partial}{\partial x}\sin\left(x^2y\right) = \cos\left(x^2y\right)\left(\frac{\partial}{\partial x}\left(x^2y\right)\right) = 2xy\cos\left(x^2y\right)$ and $f_y(x,y) = \frac{\partial}{\partial y}\sin\left(x^2y\right) = \cos\left(x^2y\right)\left(\frac{\partial}{\partial y}\left(x^2y\right)\right) = x^2\cos\left(x^2y\right)$
5. $f_x(x,y) = \frac{\partial}{\partial x}\ln\left(x^y\right) = \frac{\partial}{\partial x}\left(y\ln x\right) = \frac{y}{x}$ and $f_y(x,y) = \frac{\partial}{\partial y}\left(y\ln x\right) = \ln x$

## Answer to Exercise 2 (on page 6)

1. Finding $f_x$:

$$f_x = \frac{\partial}{\partial x}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\left[\frac{\partial}{\partial x}\left(x^2-y^2\right)\right]$$

$$f_x = 2x\cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)$$

Finding $f_y$:

$$f_y = \frac{\partial}{\partial y}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\left[\frac{\partial}{\partial y}\left(x^2-y^2\right)\right]$$

$$f_y = -2y\cos\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)$$

Finding $f_z$:

$$f_z = \frac{\partial}{\partial z}\left[\sin\left(x^2-y^2\right)\cos\left(\sqrt{z}\right)\right] = \sin\left(x^2-y^2\right)\left(-\sin\sqrt{z}\right)\cdot\left(\frac{\partial}{\partial z}\sqrt{z}\right)$$

$$f_z = \frac{-\sin\left(x^2-y^2\right)\sin\left(\sqrt{z}\right)}{2\sqrt{z}}$$

2. Finding $q_t$:

$$q_t = \frac{\partial}{\partial t}\sqrt[3]{t^3 + u^3\sin(5v)} = \frac{1}{3\left(t^3 + u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial t}\left(t^3 + u^3\sin(5v)\right)\right)$$

$$q_t = \frac{t^2}{\left(t^3 + u^3\sin(5v)\right)^{2/3}}$$

Finding $q_u$:

$$q_u = \frac{\partial}{\partial u}\sqrt[3]{t^3 + u^3\sin(5v)} = \frac{1}{3\left(t^3 + u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial u}\left(t^3 + u^3\sin(5v)\right)\right)$$

$$q_u = \frac{u^2\sin(5v)}{\left(t^3 + u^3\sin(5v)\right)^{2/3}}$$

Finding $q_v$:

$$q_v = \frac{\partial}{\partial v}\sqrt[3]{t^3 + u^3\sin(5v)} = \frac{1}{3\left(t^3 + u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial v}\left(t^3 + u^3\sin(5v)\right)\right)$$

$$q_v = \frac{u^3\cos(5v)}{3\left(t^3 + u^3\sin(5v)\right)^{2/3}}\left(\frac{\partial}{\partial v}(5v)\right) = \frac{5u^3\cos(5v)}{3\left(t^3 + u^3\sin(5v)\right)^{2/3}}$$

3. Finding $w_x$:

$$w_x = \frac{\partial}{\partial x}\left(x^z y^x\right) = \left(x^z\right)\cdot\left(\frac{\partial}{\partial x}y^x\right) + \left(y^x\right)\cdot\left(\frac{\partial}{\partial x}x^z\right)$$

$$w_x = \left(x^z\right)\left(\ln(y)y^x\right) + \left(y^x\right)\left(zx^{z-1}\right) = \left(x^{z-1}y^x\right)\left(x\ln(y) + z\right)$$

Finding $w_y$:

$$w_y = \frac{\partial}{\partial y}\left(x^z y^x\right) = \left(x^z\right)\left(\frac{\partial}{\partial y}y^x\right) = x^z\left(xy^{x-1}\right)$$

$$w_y = x^{z+1}y^{x-1}$$

Finding $w_z$:

$$w_z = \frac{\partial}{\partial z}\left(x^z y^x\right) = \left(y^x\right)\left(\frac{\partial}{\partial z}x^z\right) = \left(y^x\right)\left(\ln(x)x^z\right)$$

$$w_z = \ln(x)y^x x^z$$

## Answer to Exercise 3 (on page 8)

1. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(e^{2xy}\sin x\right)\right] = \frac{\partial}{\partial y}\left[\left(e^{2xy}\right)\left(\frac{\partial}{\partial x}\sin x\right) + (\sin x)\left(\frac{\partial}{\partial x}e^{2xy}\right)\right] = \frac{\partial}{\partial y}\left[e^{2xy}\cos x + 2ye^{2xy}\sin x\right] = \frac{\partial}{\partial y}\left(e^{2xy}\cos x\right) + \frac{\partial}{\partial y}\left(2ye^{2xy}\sin x\right) = 2xe^{2xy}\cos x + (2y)\left(\frac{\partial}{\partial y}e^{2xy}\sin x\right) + \left(e^{2xy}\sin x\right)\left(\frac{\partial}{\partial y}2y\right) = 2xe^{2xy}\cos x + 4xye^{2xy}\sin x + 2e^{2xy}\sin x$

   $f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(e^{2xy}\sin x\right)\right] = \frac{\partial}{\partial x}\left(2xe^{2xy}\sin x\right) = (2x)\left[\frac{\partial}{\partial x}\left(e^{2xy}\sin x\right)\right] + \left(e^{2xy}\sin x\right)\left(\frac{\partial}{\partial x}2x\right) = (2x)\left[\left(e^{2xy}\right)\left(\frac{\partial}{\partial x}\sin x\right) + (\sin x)\left(\frac{\partial}{\partial x}e^{2xy}\right)\right] + 2e^{2xy}\sin x = 2xe^{2xy}\cos x + 4xye^{2xy}\sin x + 2e^{2xy}\sin x = f_{xy}$

2. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(\frac{x^2}{x+y}\right)\right] = \frac{\partial}{\partial y}\left[\frac{(x+y)(2x)-x^2(1)}{(x+y)^2}\right] = \frac{\partial}{\partial y}\left[\frac{x^2+2xy}{(x+y)^2}\right] = \frac{(x+y)^2(2x)-\left(x^2+2xy\right)(2(x+y))}{(x+y)^4} = \frac{\left(x^2+2xy+y^2\right)(2x)-\left(x^2+2xy\right)(2x+2y)}{(x+y)^4} = \frac{2x^3+4x^2y+2xy^2-2x^3-2x^2y-4x^2y-4xy^2}{(x+y)^4} = \frac{-2x^2y-2xy^2}{(x+y)^4}$

   $f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(\frac{x^2}{x+y}\right)\right] = \frac{\partial}{\partial x}\left[\frac{-x^2}{(x+y)^2}\right] = \frac{(x+y)^2(-2x)-\left(-x^2\right)(2(x+y))}{(x+y)^4} = \frac{\left(x^2+2xy+y^2\right)(-2x)+x^2(2x+2y)}{(x+y)^4} = \frac{-2x^3-4x^2y-2xy^2+2x^3+2x^2y}{(x+y)^4} = \frac{-2x^2y-2xy^2}{(x+y)^4} = f_{xy}$

3. $f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(\ln(2x+3y)\right)\right] = \frac{\partial}{\partial y}\left[\frac{2}{2x+3y}\right] = \frac{-2(3)}{(2x+3y)^2} = \frac{-6}{(2x+3y)^2}$

   $f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial y}\left(\ln(2x+3y)\right)\right] = \frac{\partial}{\partial x}\left(\frac{3}{2x+3y}\right) = \frac{-3(2)}{(2x+3y)^2} = \frac{-6}{(2x+3y)} = f_{xy}$

## Answer to Exercise 4 (on page 9)

1. $f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^5y^2-3x^3y^2\right)\right] = \frac{\partial}{\partial x}\left(5x^4y^2-9x^2y^2\right) = 20x^3y^2-18xy^2.$

   $f_{xy} = f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial x}\left(x^5y^2-3x^3y^2\right)\right] = \frac{\partial}{\partial y}\left(5x^4y^2-9x^2y^2\right) = 10x^4y - 18x^2y.$

   $f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}f(x,y)\right) = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^5y^2-3x^3y^2\right)\right] = \frac{\partial}{\partial y}\left(2x^5y-6x^3y\right) = 2x^5-6x^3.$

2. $v_{pp} = \frac{\partial}{\partial p}\left(\frac{\partial}{\partial p}v(p,q)\right) = \frac{\partial}{\partial p}\left[\frac{\partial}{\partial p}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial p}\left(\cos\left(p^3+q^2\right)\left(3p^2\right)\right) = \cos\left(p^3+q^2\right)\cdot\frac{\partial}{\partial p}\left(3p^2\right) + 3p^2\cdot\frac{\partial}{\partial p}\left(\cos\left(p^3+q^2\right)\right)$

   $v_{pq} = v_{qp} = \frac{\partial}{\partial q}\left(\frac{\partial}{\partial p}v(p,q)\right) = \frac{\partial}{\partial q}\left[\frac{\partial}{\partial p}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial q}\left(\cos\left(p^3+q^2\right)\left(3p^2\right)\right) = \cos\left(p^3+q^2\right)\frac{\partial}{\partial q}\left(3p^2\right) + 3p^2\frac{\partial}{\partial q}\cos\left(p^3+q^2\right) = 0 + 3p^2\left(-\sin\left(p^3+q^2\right)\right)\left(\frac{\partial}{\partial q}\left(p^3+q^2\right)\right) = -6p^2q\sin\left(p^3+q^2\right)$

   $v_{qq} = \frac{\partial}{\partial q}\left(\frac{\partial}{\partial q}v(p,q)\right) = \frac{\partial}{\partial q}\left[\frac{\partial}{\partial q}\left(\sin\left(p^3+q^2\right)\right)\right] = \frac{\partial}{\partial q}\left[2q\cos\left(p^3+q^2\right)\right] = 2q\left[\frac{\partial}{\partial q}\cos\left(p^3+q^2\right)\right] + \cos\left(p^3+q^2\right)\left[\frac{\partial}{\partial q}(2q)\right] = (2q)\cdot\left[-2q\sin\left(p^3+q^2\right)\right] + 2\cos\left(p^3+q^2\right) = 2\cos\left(p^3+q^2\right) - 4q^2\sin\left(p^3+q^2\right)$

3. $T_{rr} = \frac{\partial}{\partial r}\left(\frac{\partial}{\partial r}T(r,\theta)\right) = \frac{\partial}{\partial r}\left[\frac{\partial}{\partial r}\left(e^{-3r}\cos\theta^2\right)\right] = \frac{\partial}{\partial r}\left(-3e^{-3r}\cos\theta^2\right) = 9e^{-3r}\cos\theta^2$
$T_{\theta r} = T_{r\theta} = \frac{\partial}{\partial \theta}\left(\frac{\partial}{\partial r}T(r,\theta)\right) = \frac{\partial}{\partial \theta}\left[-3e^{-3r}\cos\theta^2\right] = 3re^{-3r}\sin\theta^2\left(\frac{\partial}{\partial\theta}\theta^2\right) = 6r\theta e^{-3r}\sin\theta^2$
$T_{\theta\theta} = \frac{\partial}{\partial \theta}\left(\frac{\partial}{\partial \theta}T(r,\theta)\right) = \frac{\partial}{\partial \theta}\left[\frac{\partial}{\partial \theta}\left(e^{-3r}\cos\theta^2\right)\right] = \frac{\partial}{\partial \theta}\left[-e^{-3r}\sin\theta^2\left(\frac{\partial}{\partial\theta}\theta^2\right)\right] = \frac{\partial}{\partial\theta}\left(-2\theta e^{-3r}\sin\theta^2\right) = \left(-2\theta e^{-3r}\right)\left(\frac{\partial}{\partial\theta}\sin\theta^2\right) + \left(\sin\theta^2\right)\left[\frac{\partial}{\partial\theta}\left(-2\theta e^{-3r}\right)\right] = \left(-2\theta e^{-3r}\right)\left(\cos\theta^2\right)\left(\frac{\partial}{\partial\theta}\theta^2\right) + \left(\sin\theta^2\right)\left(-2e^{-3r}\right) = -4\theta^2 e^{-3r}\cos\theta^2 - 2e^{-3r}\sin\theta^2$

## Answer to Exercise 5 (on page 12)

1. We are looking for $dz/dt$ only:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}[\sin x \cos y]\cdot\frac{d}{dt}\left[3\sqrt{t}\right] + \frac{\partial}{\partial y}[\sin x\cos y]\cdot\frac{d}{dt}[2/t]$$

$$= (\cos x\cos y)\cdot\left(\frac{3}{2\sqrt{t}}\right) + (-\sin x\sin y)\cdot\left(-\frac{2}{t^2}\right)$$

$$= \frac{3\cos x\cos y}{2\sqrt{t}} + \frac{2\sin x\sin y}{t^2}$$

Substituting for $x$ and $y$:

$$\frac{dz}{dt} = \frac{3\cos\left(3\sqrt{t}\right)\cos(2/t)}{2\sqrt{t}} + \frac{2\sin\left(3\sqrt{t}\right)\sin(2/t)}{t^2}$$

2. We are looking for $dz/dt$ only:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= \frac{\partial}{\partial x}\left[\sqrt{1+xy}\right]\cdot\frac{d}{dt}[\tan t] + \frac{\partial}{\partial y}\left[\sqrt{1+xy}\right]\cdot\frac{d}{dt}[\arctan t]$$

$$= \left(\frac{y}{2\sqrt{1+xy}}\right)\cdot\left(\sec^2 t\right) + \left(\frac{x}{2\sqrt{1+xy}}\right)\cdot\left(\frac{1}{t^2+1}\right)$$

Substituting for $x$ and $y$:

$$\frac{dz}{dt} = \frac{\tan t\sec^2 t}{2\sqrt{1+\tan t\arctan t}} + \frac{\tan t}{2\sqrt{1+\tan t\arctan t}\,(t^2+1)}$$

$$= \frac{\tan t}{2\sqrt{1+\tan t\arctan t}}\left(\sec^2 t + \frac{1}{t^2+1}\right)$$

3. Finding $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left[\arctan\left(x^2+y^2\right)\right]\cdot\frac{\partial}{\partial s}\left[t\ln s\right] + \frac{\partial}{\partial y}\left[\arctan\left(x^2+y^2\right)\right]\cdot\frac{\partial}{\partial s}\left[se^t\right]$$

$$\left(\frac{2x}{(x^2+y^2)^2+1}\right)\cdot\left(\frac{t}{s}\right) + \left(\frac{2y}{(x^2+y^2)^2+1}\right)\cdot\left(e^t\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = \left(\frac{2\left(t\ln s\right)}{\left[(t\ln s)^2+(se^t)^2\right]^2+1}\right)\cdot\left(\frac{t}{s}\right) + \left(\frac{2\left(se^t\right)}{\left[(t\ln s)^2+(t\ln s)^2\right]^2+1}\right)\cdot\left(e^t\right)$$

$$= \left(\frac{2}{\left[t^2(\ln s)^2+se^{2t}\right]^2+1}\right)\cdot\left(\frac{t^2\ln s}{s}+se^{2t}\right)$$

Finding $\partial z/\partial t$:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{\partial}{\partial x}\left(\arctan\left(x^2+y^2\right)\right)\cdot\frac{\partial}{\partial t}\left(t\ln s\right) + \frac{\partial}{\partial y}\left(\arctan\left(x^2+y^2\right)\right)\cdot\frac{\partial}{\partial t}\left(se^t\right)$$

$$= \left(\frac{2x}{(x^2+y^2)^2+1}\right)\cdot\left(\ln s\right) + \left(\frac{2y}{(x^2+y^2)^2+1}\right)\cdot\left(se^t\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = \left(\frac{2}{\left[(t\ln s)^2+(se^t)^2\right]^2+1}\right)\cdot\left[t(\ln s)^2+s^2e^{2t}\right]$$

4. Finding $\partial z/\partial s$:

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x}\left(\sqrt{x}e^{xy}\right)\cdot\frac{\partial}{\partial s}\left(1+st\right) + \frac{\partial}{\partial y}\left(\sqrt{x}e^{xy}\right)\cdot\frac{\partial}{\partial s}\left(s^2-t^2\right)$$

$$= \left[\frac{e^{xy}\left(2xy+1\right)}{2\sqrt{x}}\right]\cdot\left(t\right) + \left[x^{3/2}e^{xy}\right]\cdot\left(2s\right)$$

$$= \left[\frac{e^{xy}}{\sqrt{x}}\right]\cdot\left(\frac{\left(2xy+1\right)t}{2}+x^2\left(2s\right)\right)$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial s} = \left[\frac{e^{(1+st)(s^2-t^2)}}{\sqrt{1+st}}\right] \cdot \left(\frac{\left(2(1+st)(s^2-t^2)+1\right)t}{2} + (1+st)^2(2s)\right)$$

Finding $\partial z/\partial t$:

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \\ &= \frac{\partial}{\partial y}\left[\sqrt{x}e^{xy}\right] \cdot \frac{\partial}{\partial t}[1+st] + \frac{\partial}{\partial y}\left[\sqrt{x}e^{xy}\right] \cdot \frac{\partial}{\partial t}\left[s^2-t^2\right] \\ &= \left[\frac{e^{xy}(2xy+1)}{2\sqrt{x}}\right] \cdot (s) + \left[x^{3/2}e^{xy}\right] \cdot (-2t) \\ &= \left[\frac{e^{xy}}{\sqrt{x}}\right] \cdot \left[\frac{(2xy+1)s}{2} - 2tx^2\right]\end{aligned}$$

Substituting for $x$ and $y$:

$$\frac{\partial z}{\partial t} = \left[\frac{e^{(1+st)(s^2-t^2)}}{\sqrt{(1+st)}}\right] \cdot \left[\frac{\left(2(1+st)\left(s^2-t^2\right)+1\right)s}{2} - 2t(1+st)^2\right]$$

## Answer to Exercise 6 (on page 17)

1. $z(1,-1) = (1)^2e(-1/1 = 1/e$. Therefore, we are looking for tangent lines through the point $(1,-1,1/e)$. Finding a tangent line parallel to the x-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left(x^2e^{y/x}\right) = x^2\left(\frac{\partial}{\partial x}e^{y/x}\right)+e^{y/x}\left(\frac{\partial}{\partial x}x^2\right) = x^2e^{y/x}\left(\frac{\partial}{\partial x}\frac{y}{x}\right)+2xe^{y/x} = x^2e^{y/x}\left(\frac{-y}{x^2}\right)+2xe^{y/x} = (2x-y)e^{y/x}$ and $z_x(1,-1) = (2(1)-(-1))e^{-1/1} = (3)e^{-1} = 3/e$. So, the slope of a line tangent to the surface at $(1,-1,1/e)$ parallel to the x-axis is $3/e$ and an equation for that line is $z = 3/e(x-1)-1/e$.

   Finding a tangent line parallel to the y-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left(x^2e^{y/x}\right) = xe^{y/x}$ and $z_y(1,-1) = (1)e^{-1/1} = 1/e$. So, the slope of a line tangent to the surface at $(1,-1,1/e)$ parallel to the y-axis is $1/e$ and an equation for that line is $z = 1/e(y+1)-1/e$.

   The function is changing faster in the x-direction.

2. $z(\pi,\pi/2) = \cos(\pi)+\frac{\pi}{2}\sin(\pi/2) = \frac{\pi}{2}-1$. Therefore, we are looking for tangent lines through the point $(\pi,\pi/2,\pi/2-1)$. Finding a tangent line parallel to the x-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(\cos x + y\sin y) = -\sin x$ and $z_x(\pi,\pi/2) = -\sin\pi = 0$. So, the slope of a line tangent to the surface at $(\pi,\pi/2,\pi/2-1)$ parallel to the x-axis is $0$ and an equation for that line is $z = \pi/2-1$.

   Finding a tangent line parallel to the y-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(\cos x + y\sin y) = y\left(\frac{\partial}{\partial y}\sin y\right)+\sin y\left(\frac{\partial}{\partial y}y\right) = y\cos y+\sin y$ and $z_y(\pi,\pi/2) = \left(\frac{\pi}{2}\right)\cos\left(\frac{\pi}{2}\right)+\sin\left(\frac{\pi}{2}\right) = 1$. So, the slope

of a line tangent to the surface at $(\pi, \pi/2, \pi/2-1)$ parallel to the $y$-axis is 1 and an equation for that line is $z = (y - \pi/2) - (\pi/2 - 1) = y - \pi + 1$.

The function is changing faster in the $y$-direction.

3. $z(3,2) = 3^2(2) - 3(3)(2^2) = 18 - 36 = -18$. Therefore, we are looking for tangent lines through the point $(3, 2, -18)$. Finding a tangent line parallel to the $x$-axis: $\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left(x^2y - 3xy^2\right) = 2xy - 3y^2$ and $z_x(3,2) = 2(3)(2) - 3(2)^2 = 0$. So, the slope of a line tangent to the surface at $(3, 2, -18)$ is 0 and an equation for that line is $z = -18$

   Finding a tangent line parallel to the $y$-axis: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left(x^2y - 3xy^2\right) = x^2 - 6xy$ and $z_y(3,2) = 3^2 - 6(3)(2) = 9 - 36 = -27$. So, the slope of a line tangent to the surface at $(3, 2, -18)$ is $-27$ and an equation for that line is $z = -27(y-2) + -18 = -27y + 54 - 18 = 36 - 27y$.

   The function is changing faster in the $y$-direction.

## Answer to Exercise 7 (on page 25)

1. First, we define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [-3k, -4k]$$

$$\sqrt{(-3k)^2 + (4k)^2} = 1$$

$$\sqrt{9k^2 + 16k^2} = \sqrt{25k^2} = 5k = 1$$

$$k = \frac{1}{5}$$

   Therefore, we define $\mathbf{u} = [-3/5, 4/5]$ and the directional derivative is given by:

$$D_u(x,y) = \left(\frac{-3}{5}\right)\frac{\partial}{\partial x}f(x,y) + \left(\frac{4}{5}\right)\frac{\partial}{\partial y}f(x,y)$$

$$= \left(\frac{-3}{5}\right)\frac{\partial}{\partial x}\left[e^{3x}\sin 2y\right] + \left(\frac{4}{5}\right)\frac{\partial}{\partial y}\left[e^{3x}\sin 2y\right]$$

$$= \left(\frac{-3}{5}\right)\left(3e^{3x}\sin 2y\right) + \left(\frac{4}{5}\right)\left(2e^{3x}\cos 2y\right)$$

   And substituting for $(x,y) = (0, \pi/6)$:

$$D_u(0,\pi/6) = \left(\frac{-3}{5}\right)\cdot\left[3e^{3\cdot 0}\sin\left(\frac{\pi}{3}\right)\right] + \left(\frac{4}{5}\right)\cdot\left[2e^{3\cdot 0}\cos\left(\frac{\pi}{3}\right)\right]$$

$$D_u(0,\pi/6) = \left(\frac{-3}{5}\right)\cdot\left[3\cdot\frac{\sqrt{3}}{2}\right] + \left(\frac{4}{5}\right)\cdot\left[2\cdot\frac{1}{2}\right]$$

$$d_u(0,\pi/6) = \left(\frac{-3}{5}\right)\cdot\left(\frac{3\sqrt{3}}{2}\right) + \left(\frac{4}{5}\right)\cdot(1)$$

$$D_u(0,\pi/6) = \frac{-9\sqrt{3}}{10} + \frac{8}{10} = \frac{8-9\sqrt{3}}{10} \approx -0.759$$

2. We can express $\mathbf{v}$ as $\mathbf{v} = [2,-1]$. And we define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [2k,-k]$$

$$\sqrt{(2k)^2 + (-k)^2} = 1$$

$$\sqrt{4k^2 + k^2} = \sqrt{5}k = 1$$

$$k = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

Therefore, we define $\mathbf{u} = \left[2\sqrt{5}/5, -\sqrt{5}/5\right]$ and the directional derivative is given by:

$$D_u(x,y) = \left(\frac{2\sqrt{5}}{5}\right)\frac{\partial}{\partial x}f(x,y) + \left(\frac{-\sqrt{5}}{5}\right)\frac{\partial}{\partial y}f(x,y)$$

$$= \left(\frac{2\sqrt{5}}{5}\right)\frac{\partial}{\partial x}\left[x^2y + xy^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\frac{\partial}{\partial y}\left[x^2y + xy^3\right]$$

$$= \left(\frac{2\sqrt{5}}{5}\right)\left[2xy + y^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\left[x^2 + 3xy^2\right]$$

And substituting $(x,y) = (2,4)$:

$$D_u(2,4) = \left(\frac{2\sqrt{5}}{5}\right)\left[2(2)(4) + 4^3\right] + \left(\frac{-\sqrt{5}}{5}\right)\left[2^2 + 3(2)(4^2)\right]$$

$$D_u(2,4) = \left(\frac{2\sqrt{5}}{5}\right)[80] + \left(\frac{-\sqrt{5}}{5}\right)[100]$$

$$D_u(2,4) = 32\sqrt{5} - 20\sqrt{5} = 12\sqrt{5} \approx 26.833$$

3. We define $\mathbf{u}$ such that $|\mathbf{u}| = 1$ and $\mathbf{u}$ is in the same direction as $\mathbf{v}$:

$$\mathbf{u} = k\mathbf{v} = [k,k,k]$$

$$\sqrt{k^2 + k^2 + k^2} = 1$$

$$\sqrt{3}k = 1$$

$$k = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

Therefore, we let $\mathbf{u} = \left[\sqrt{3}/3, \sqrt{3}/3, \sqrt{3}/3\right]$ and the directional derivative is given by:

$$D_u(x,y,z) = \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial x}f(x,y,z) + \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial y}f(x,y,z) + \left(\frac{\sqrt{3}}{3}\right)\frac{\partial}{\partial z}f(x,y,z)$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{\partial}{\partial x}\ln\left(x^2+3y-z\right) + \frac{\partial}{\partial y}\ln\left(x^2+3y-z\right) + \frac{\partial}{\partial z}\ln\left(x^2+3y-z\right)\right]$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{2x}{x^2+3y-z} + \frac{3}{x^2+3y-z} + \frac{-1}{x^2+3y-z}\right]$$

$$= \left(\frac{\sqrt{3}}{3}\right)\left[\frac{2x+2}{x^2+3y-z}\right] = \frac{\sqrt{3}\,(2x+2)}{3\,(x^2+3y-z)}$$

And substituting $(x,y,z) = (2,2,1)$:

$$D_u(2,2,1) = \frac{\sqrt{3}\,(2(2)+2)}{3\,(2^2+3(2)-1)} = \frac{\sqrt{3}\,(6)}{3\,(9)} = \frac{2\sqrt{3}}{9} \approx 0.385$$

## Answer to Exercise 8 (on page 28)

1. $z = f(50,50) = 3000 - 0.01(50)^2 - 0.02(50)^2 = 2925$
2. A south-pointing unit vector is $\mathbf{u} = [0,-1]$. To find the rate of change, we find the directional derivative in the direction of $\mathbf{u}$ at $(50,50)$:

$$D_u f(x,y) = (-1)\left[\frac{\partial}{\partial y}\left(3000 - 0.01x^2 - 0.02y^2\right)\right]$$

$$D_u f(x,y) = (-1)\,(-0.04y) = 0.04y$$

   And at $(50,50)$, $D_u f(50,50) = 0.04(50) = 2 > 0$. Therefore, if you walk south, you will ascend.
3. A northwest-pointing unit vector is $\mathbf{u} = \left[-\sqrt{2}/2, \sqrt{2}/2\right]$. To find the rate of change, we find the directional derivative at $(50,50)$ in the direction of $\mathbf{u}$:

$$D_u f(x,y) = \left(\frac{-\sqrt{2}}{2}\right)\left[\frac{\partial}{\partial x}f(x,y)\right] + \left(\frac{\sqrt{2}}{2}\right)\left[\frac{\partial}{\partial y}f(x,y)\right]$$

$$D_u f(x,y) = \left(\frac{-\sqrt{2}}{2}\right)[-0.02x] + \left(\frac{\sqrt{2}}{2}\right)[-0.04y]$$

$$D_u f(x,y) = 0.01\sqrt{2}x - 0.02\sqrt{2}y$$

$$D_u f(50,50) = 0.01\sqrt{2}\,(50) - 0.02\sqrt{2}\,(50) = \frac{-\sqrt{2}}{2} \approx -0.707$$

The rate of elevation change walking northwest is approximately $-0.707$, so you will descend and your rate of elevation change would be less than if you walked south.

4. To find the direction of maximum elevation gain, we find the direction the gradient vector points in:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$
$$\nabla f = [-0.02x, -0.04y]$$

And at $(50,50)$,

$$\nabla f(50,50) = [-0.02(50), -0.04(50)] = [-1,-2]$$

Therefore, the rate of greatest elevation change is in a south-by-southwest direction indicated by the vector $[-1,-2]$ and the rate of elevation change is $|\nabla f(50,50)| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$. Notice this is greater than the other two rates of change we have found.

## Answer to Exercise 9 (on page 30)

1.

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(x^2 - y^2\right)\right] + \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(x^2 - y^2\right)\right]$$
$$= \frac{\partial}{\partial x}[2x] + \frac{\partial}{\partial y}[-2y] = 2 - 2 = 0$$

Therefore, $f(x,y) = x^2 - y^2$ is a solution to Laplace's Equation.

2.

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}(\sin x \cosh y + \cos x \sinh y)\right] + \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}(\sin x \cosh y + \cos x \sinh y)\right]$$
$$= \frac{\partial}{\partial x}[\cos x \cosh y - \sin x \sinh y] + \frac{\partial}{\partial y}[\sin x \sinh y + \cos x \cosh y]$$
$$= -\sin x \cosh y - \cos x \sinh y + \sin x \cosh y + \cos x \sinh y = 0$$

Therefore, $f(x,y) = \sin x \cosh y + \cos x \sinh y$ is a solution to Laplace's Equation.

3.

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(e^{-x}\cos y - e^{-y}\cos x\right)\right] + \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y}\left(e^{-x}\cos y - e^{-y}\cos x\right)\right]$$

$$= \frac{\partial}{\partial x}\left[-e^{-x}\cos y + e^{-y}\sin x\right] + \frac{\partial}{\partial y}\left[-e^{-x}\sin y + e^{-y}\cos x\right]$$
$$= e^{-x}\cos y + e^{-y}\cos x - e^{-x}\cos y - e^{-y}\cos x = 0$$

Therefore, $f(x,y) = e^{-x}\cos y - e^{-y}\cos x$.

## Answer to Exercise 10 (on page 32)

1. Finding the partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}\left(\cos(kx)\cos(akt)\right)\right] = \frac{\partial}{\partial t}\left[-ak\cos(kx)\sin(akt)\right]$$

$$\frac{\partial^2 f}{\partial t^2} = -a^2k^2\cos(kx)\cos(akt)$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\cos(kx)\cos(akt)\right)\right] = \frac{\partial}{\partial x}\left[-k\sin(kx)\cos(akt)\right]$$

$$\frac{\partial^2 f}{\partial x^2} = -k^2\cos(kx)\cos(akt)$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = -a^2k^2\cos(kx)\cos(akt) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \cos(kx)\cos(akt)$ satisfies the Wave Equation.

2. Finding the partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}\left(\sin(x-at) + \ln(x+at)\right)\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[-a\cos(x-at) + \frac{a}{x+at}\right] = -a^2\sin(x-at) + \frac{-a^2}{(x+at)^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\sin(x-at) + \ln(x+at)\right)\right]$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\cos(x-at) + \frac{1}{x+at}\right] = -\sin(x-at) + \frac{-1}{(x+at)^2}$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = -a^2\sin(x-at) + \frac{-a^2}{(x+at)^2} = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \sin(x-at) + \ln(x+at)$ satisfies the Wave Equation.

3. Finding the partial derivatives:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial}{\partial t}\left(\frac{t}{a^2t^2 - x^2}\right)\right] = \frac{\partial}{\partial t}\left[\frac{\left(a^2t^2 - x^2\right) - t\left(2a^2t\right)}{\left(a^2t^2 - x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{a^2t^2 - x^2 - 2a^2t^2}{\left(a^2t^2 - x^2\right)^2}\right] = \frac{\partial}{\partial t}\left[\frac{-a^2t^2 - x^2}{\left(a^2t^2 - x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\left(a^2t^2 - x^2\right)^2\left(-2a^2t\right) - \left(-a^2t^2 - x^2\right)\left(2\left(a^2t^2 - x^2\right)\left(2a^2t\right)\right)}{\left(a^2t^2 - x^2\right)^4}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{\left(a^2t^2 - x^2\right)\left(-2a^2t\right) - \left(-a^2t^2 - x^2\right)\left(4a^2t\right)}{\left(a^2t^2 - x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{-2a^4t^3 + 2a^2tx^2 + 4a^4t^3 + 4a^2tx^2}{\left(a^2t^2 - x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{2a^4t^3 + 6a^2tx^2}{\left(a^2t^2 - x^2\right)^3} = 2a^2t\left(\frac{a^2t^2 + 3x^2}{\left(a^2t^2 - x^2\right)^3}\right)$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left(\frac{t}{a^2t^2 - x^2}\right)\right] = \frac{\partial}{\partial x}\left[\frac{\left(a^2t^2 - x^2\right)(0) - t(-2x)}{\left(a^2t^2 - x^2\right)^2}\right]$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{2tx}{\left(a^2t^2 - x^2\right)^2}\right] = \frac{\left(a^2t^2 - x^2\right)^2(2t) - (2tx)\left(2\left(a^2t^2 - x^2\right)(-2x)\right)}{\left(a^2t^2 - x^2\right)^4}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\left(a^2t^2 - x^2\right)(2t) - (2tx)(2)(-2x)}{\left(a^2t^2 - x^2\right)^3}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2a^2t^3 - 2tx^2 + 8tx^2}{\left(a^2t^2 - x^2\right)^3} = \frac{2a^2t^3 + 6tx^2}{\left(a^2t^2 - x^2\right)^3} = 2t\left(\frac{a^2t^2 + 3x^2}{\left(a^2t^2 - x^2\right)^3}\right)$$

And we see that:

$$a^2\frac{\partial^2 f}{\partial x^2} = 2a^2t\left(\frac{a^2t^2 + 3x^2}{\left(a^2t^2 - x^2\right)^3}\right) = \frac{\partial^2 f}{\partial t^2}$$

Therefore, $f(x,t) = \frac{t}{a^2t^2 - x^2}$ satisfies the Wave Equation.

## Answer to Exercise 11 (on page 34)

1. The marginal utility of labor is given by $\partial P/\partial L$:

$$\frac{\partial P}{\partial L} = \frac{\partial}{\partial L}\left[1.01 L^{0.75} K^{0.25}\right] = 0.7575 L^{-0.25} K^{0.25}$$

2. The marginal utility of capital is given by $\partial P/\partial K$:

$$\frac{\partial P}{\partial K} = \frac{\partial}{\partial K}\left[1.01 L^{0.75} K^{0.25}\right] = 0.2525 L^{0.75} K^{-0.75}$$

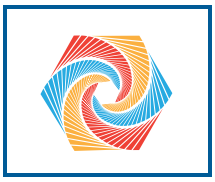
3. Finding the marginal utility of labor in 1916:

$$\frac{\partial P}{\partial L} = 0.7575\,(382)^{-.25}\,(126)^{0.25} \approx 0.574$$

   And finding the marginal utility of capital in 1916:

$$\frac{\partial P}{\partial K} = 0.2525\,(382)^{0.75}\,(126)^{-0.75} \approx 0.580$$

4. Since the marginal utility of capital is greater, I would invest in capital. This would yield a greater increase in production than the same investment in labor.



# Index

Clairaut's theorem, 7

directional derivative, 17

gradient, 1, 26

Laplace's equation, 29

partial derivative, 1
    chain rule for, 9
    limits of, 2